VOICECARE AI

Safe Growth Operating System

Workflow and Technical Architecture Guide

Sequential Multi-Agent Framework for Safe Brand Scaling

A technical deep dive into the 6-agent sequential pipeline engineered to accelerate LinkedIn follower growth, capture category share-of-voice, and optimize AI engine visibility (GEO/AEO/ASO) while adhering to strict healthcare compliance (PHI protection) and platform anti-spam boundaries.

Document Scope:	Prototype & Architecture Specification
Release Version:	v1.3.4
Target Follower Goal:	2,900 to 10,000 Followers in 90 Days (~590 net-new/week)
Core Technologies:	HTML5, CSS3, ES Modules, Vercel Serverless, Groq API
Safety Posture:	**100% Pure Research & Drafting (Human-in-the-Loop Required)**

1. Executive Summary

The VoiceCare AI Safe Growth Engine is an operating console designed to scale B2B marketing and brand visibility. Traditionally, rapid social scaling relies on automated bots, scrapers, or bought leads - tactics that lead to severe platform penalties, profile bans, and brand erosion. In healthcare, these risks are compounded by the necessity to safeguard patient data and protect private health histories under strict HIPAA frameworks.

This platform acts as an administrative cockpit. It utilizes a sequential chain of six domain-specific AI sub-agents to analyze competitors, draft tailored content, audit SEO blogs, calculate growth velocities, compile third-party authority citation lists, and map assets to conversion funnels. The core architectural boundary is simple: the AI researches and drafts, but never executes external API actions. Humans review, verify, and post everything manually.

2. Technical Stack Architecture

The application is structured to minimize client-side dependencies, reduce payload weights, and securely isolate sensitive credentials. The stack components are:

- **HTML5 UI:** A single-page, responsive console layout utilizing semantic elements. The workspace is divided into a Company Profile input panel, an Orchestration Control bar (for single-agent or full sprint execution), and a grid of six agent output cards.
- **Vanilla CSS3:** A custom, responsive design system engineered using CSS Grid and Flexbox. Features CSS variables for light/dark themes, scroll-safe code blocks, and dynamic agent state status lights.
- **Vanilla JS & ES Modules:** Handles form state retrieval, coordinates DOM renders, parses JSON payloads, and runs sequential orchestration pipelines using browser-native async fetch handlers (no Webpack or Babel required).
- **Vercel Serverless Function:** All model routing and data gathering run securely inside `/api/agent.js`. Groq, OpenRouter, and Jina keys are kept strictly on Vercel's server environment, preventing credential leakage to the browser.
- **Jina AI Web Grounding:** Used for real-time internet data gathering. Jina Search (`s.jina.ai`) queries public indexers and returns clean results, while Jina Reader (`r.jina.ai`) scrapes raw body copy from blog URLs and source assets.
- **Groq & OpenRouter APIs:** High-velocity LLM inference. Under rate limits or Groq service downtime, the backend automatically routes requests to OpenRouter equivalents (e.g., google/gemini-2.5-flash) via a multi-model fallback chain.

IMPORTANT DEPLOYMENT NOTE

To prevent Vercel Hobby plan 10-second timeouts, the backend runs dynamic time budgeting (9.5s max total cap) and caps Jina crawls/searches to 1 per request. Running on localhost bypasses all timeout budgets and concurrency limits to retrieve maximum competitor details.

3. Sequential Multi-Agent Workflow

A key design pattern of the Safe Growth Engine is context propagation. Rather than treating each agent as an isolated query, the system runs a sequential sprint where findings from upstream intelligence agents are structured and forwarded into downstream creative and planning agents.

CONTEXT PROPAGATION FLOW CHART

```
[Input Profile] -> AGENT 01 (Market Intelligence) -> Extracts Gaps & Narrative
                                                         |
           +---------------------------------------------+
           v                                             v
AGENT 02 (LinkedIn Content)                     AGENT 03 (GEO Blog Audit)
Outputs: Founder & Page Posts                   Outputs: Audits & Content Briefs
           |                                             |
           +----------------------+----------------------+
                                  v
[Agents 04, 05, 06] -> Weekly Forecasting, Citation Building, & Conversion Mapping
```

In a Full Sprint execution, the browser coordinates this sequence automatically:

- **Step 1: Market Analysis:** Agent 01 (Market Intelligence) executes first. It reads the company profile and competitors, compiles a competitive snapshot and a Share-of-Voice (SOV) scorecard, and identifies the Top 5 Content Gaps.
- **Step 2: Context Extraction:** The JavaScript orchestrator extracts the generated 'Top 5 Content & Category Gaps' and 'Category Narrative' sections from Agent 01's response using clean header boundary parsers.
- **Step 3: Creative & SEO Directives:** The extracted content is injected as competitive context variables into the payloads for Agent 02 (LinkedIn Content) and Agent 03 (GEO Visibility), ensuring the generated post drafts and blog briefs directly target competitor weaknesses.
- **Step 4: Continuous Pipelines:** The remaining agents (Agent 04 Growth, Agent 05 Citations, Agent 06 Conversion) consume metrics, directory lists, and source assets to complete the comprehensive growth sprint pack, which is saved as a single unified document.

Targeting & Audience Personas:

- **Ideal Customer Profile (ICP):** Configures the company's B2B target market segment. Upstream and downstream agents use this profile to filter directory recommendations and align search terms.
- **Target Persona:** Defines the job role or executive title targeted (e.g. CFOs, Practice Managers). Post drafts, carousel outlines, and hook angles are written to address this audience directly.
- **Brand Tone:** Enforces verbal constraints (e.g. analytical, expert, direct) to ensure all drafted outputs match your company's brand identity.

Data Grounding & Anti-Hallucination Guardrails:

- **Parallel Search Gathering:** To satisfy Vercel's strict 10s budget, the backend triggers Jina searches concurrently. In Market Intel, Jina runs parallel queries for each competitor (e.g. 'competitor linkedin content topics OR authority news OR citation directories') alongside the main category query. This gathers complete grounding data in a single turn before LLM execution.
- **Hallucination Prevention:** If specific details are missing from Jina's search results (such as LinkedIn content pillars or citation sources), hardened system instructions force the LLM to output 'Not found in search' or 'Not publicly indexed' rather than fabricating mock placeholder figures.

4. Detailed Agent Profiles

Each agent runs under strict operational constraints defined in its system prompt configuration. The tables below specify the core parameters, system behaviors, and required markdown outputs.

Agent 01 - Market Intelligence + Share of Voice

Temp: 0.4 | Max Tokens: 2200

Models:	Primary: groq/compound | Fallback: llama-3.3-70b-versatile
Behavior:	Identifies market gaps, catalogs competitor positioning, and builds SOV matrices.
Outputs:	*Competitor Snapshot, Competitor Scorecard, Top 5 Gaps, Differentiation Angles, Category Narrative, Safe Actions*

Agent 02 - LinkedIn Content + Founder Growth

Temp: 0.7 | Max Tokens: 2400

Models:	Primary: llama-3.3-70b-versatile | Fallback: llama-3.1-8b-instant
Behavior:	Drafts LinkedIn posts for company and leadership, outline slides, comments, and amplification kits.
Outputs:	*Company Posts, Founder Posts, Employee Advocacy, Carousel Outline, Comments, Amplification Kit, Repurposing Plan*

Agent 03 - AI Search / GEO-AEO-ASO Audit

Temp: 0.4 | Max Tokens: 2600

Models:	Primary: groq/compound | Fallback: llama-3.3-70b-versatile
Behavior:	Conducts existing blog audits and structures GEO content briefs to capture AI engine citations.
Outputs:	*Existing Blog Audit, Content Brief, Citation Testing Protocol, Content Gap Summary, Safe Actions*

Agent 04 - Growth Measurement + Forecasting

Temp: 0.3 | Max Tokens: 1900

Models:	Primary: llama-3.3-70b-versatile | Fallback: llama-3.1-8b-instant
Behavior:	Performs follower growth pace arithmetic and tracks visibility indices against goal pacing.
Outputs:	*Pace-to-Goal Calculation, Reverse Follower Model, Visibility Scorecard, Metrics Observations, Next Experiments*

4. Detailed Agent Profiles (Cont.)

Agent 05 - Third-Party Citation + Authority

Temp: 0.4 | Max Tokens: 2200

Models: Primary: groq/compound | Fallback: llama-3.3-70b-versatile
Behavior: Discovers authoritative third-party directories, earned media opportunities, and compliance pages.
Outputs: *Citation Opportunity List, Priority Targets, 5 Pitch Angles, Trust Asset Requirements, Safe Actions*

Agent 06 - Conversion Asset + Repurposing

Temp: 0.6 | Max Tokens: 2400

Models: Primary: llama-3.3-70b-versatile | Fallback: llama-3.1-8b-instant
Behavior: Maps user attention to conversion funnels (demos/webinars) and repurposes assets into multi-channel content.
Outputs: *Conversion Asset Map, CTA Mapping, Repurposing Matrix, Original Benchmark Asset Ideas, Safe Actions*

5. Safety Boundaries & Compliance

To ensure corporate safety, protect user accounts, and prevent regulatory issues, a shared safety core is interpolated into the system prompt of every agent. These eight rules override any user request or agent inference behavior.

SHARED COMPLIANCE BOUNDARY RULES (SAFETY_CORE)

1. NO AUTOMATION: The agent only writes drafts; it has no access to post or edit social media.
2. NO SCRAPING: The agent only uses public information or manually provided analytics.
3. NO FABRICATION: The agent never invents names, numbers, or claims; missing details use placeholders.
4. NO AUTO-ENGAGEMENT: No browser automation, proxy networks, fake accounts, or pods.
5. ZERO PHI INGESTION: The agent never accepts or references Protected Health Information.
6. COMPLIANCE GUARDRAILS: No clinical, medical, or security claims without explicit verification.
7. ASO STANDARDS: Focus ASO entirely on AI Search Optimization / Answer Search Optimization.
8. MARKDOWN ONLY: The agent must output clean, raw Markdown structures with no meta preambles.

By enforcing these guardrails at the prompt level and maintaining a strict Human-in-the-Loop checkpoint, VoiceCare AI scales its LinkedIn reach and improves search citations while keeping its digital presence 100% compliant and threat-free.